RESEARCH ARTICLE

Preventing Environmental-Claim Misrepresentation in the European Union

An Information-Safe Agent-Based Comparison of Ex-Post Supervision, Algorithmic Pre-Screening, and Certified Data Infrastructure

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Simulation-based policy experiment. Results are not empirical forecasts, causal estimates, legal advice, or evidence of real-world policy effectiveness. Legal and model cut-off: 15 July 2026.

Abstract

Environmental-claim governance combines an information problem with an institutional-capacity problem: regulators observe claims and uncertain evidence rather than firms' latent environmental states, while preventive scrutiny may suppress truthful communication. An information-safe agent-based model links corporate environmental claims to consumer demand, investor valuation, workforce responses, a limit-order-book market, and capacity-constrained supervision. The experiment compares current EU-style ex-post supervision with two proposed institutions: a voluntary algorithmic pre-screening hub and a certified green-data connector. A 28-parameter Latin-hypercube design evaluates 200 configurations at 120, 365, 1,000, and 2,000 days, with three paired common-random-number replications per configuration; a separate executed extension re-evaluates 12 representative configurations at 15 replications. Within the model, the hub reduces exposure-weighted claim severity in 85.0-90.5% of configurations across horizons, but systematically increases greenhushing and public and firm compliance costs. The connector is weak and heterogeneous at 120 days, improves severity in 60.5% of configurations at 365 days, and becomes robust at 1,000 and 2,000 days (97.0% and 100.0%), while remaining cost-intensive and increasing evidence-conflict workload. Default rankings reverse across horizons in 80.5% of matched configurations, showing why policy comparisons must be conditional on time, evidence quality, capacity, participation, and normative weights. A 30-seed stylized-facts validation of the market engine reproduces volatility clustering, positive spread and depth, trade-sign persistence, and cancellation activity; partially reproduces fat tails, the volume-volatility relation, and weak linear return autocorrelation; and does not reproduce positive price impact under the implemented diagnostic. The market layer is therefore reported as endogenous market response and financing context: firm greenwashing, disclosure, and transition-investment rules contain no own-share-price feedback, and no conclusion relies on stock-price discipline of corporate behavior. The results are transparent simulation-based policy orderings, not empirical forecasts or causal estimates.

Keywords: agent-based computational economics; greenwashing; greenhushing; environmental claims; sustainable finance; consumer protection; regulatory technology; European Union; global sensitivity analysis; information asymmetry

1. Introduction

Environmental communication is economically consequential because it can alter consumer choice, investor valuation, employee identification, and access to finance. The same communication is difficult to govern. A claim can be inaccurate because management strategically exaggerates, because the evidence is incomplete, because operational and reporting boundaries are not comparable, or because uncertainty is poorly disclosed. A public authority rarely observes the firm's physical environmental state directly. It works through claims, records, assurance procedures, complaints, public registers, and finite investigative capacity. The regulatory problem is therefore not simply to classify statements as true or false. It is to design procedures that improve the reliability of communication without confusing measurement error with misconduct or inducing firms to withdraw defensible information.

Research on greenwashing has described institutional, market, organizational, and individual drivers of misleading environmental communication (Delmas and Burbano, 2011), the diversity of practices covered by the term (Lyon and Montgomery, 2015), and the role of accusation and legitimacy in determining how claims are interpreted (Seele and Gatti, 2017). Economic work also shows that stronger scrutiny can deter exaggerated disclosure while causing some firms to disclose less overall (Lyon and Maxwell, 2011). Field evidence on third-party sustainability ratings similarly indicates that independent evaluation can deter greenwashing (Parguel, Benoit-Moreau, and Larceneux, 2011). Selective disclosure is not reducible to a single firm-level environmental score: it depends on the claim, the omitted information, the audience, and the surrounding evidence (Marquis, Toffel, and Zhou, 2016). Greenhushing - the under-communication of sustainability practices - is established in the literature (Font, Elgammal, and Lamond, 2017), while strategic silence shows how withholding favourable certification information can be used to manage scrutiny and hypocrisy risk (Carlos and Lewis, 2018).

These observations motivate a computational approach. Agent-based computational economics represents economic processes as dynamic systems generated by heterogeneous, interacting agents (Tesfatsion, 2006). Agent-based finance is especially useful when investor strategies, market microstructure, credit, and feedback between prices and balance sheets matter (LeBaron, 2006). In the present model, corporate environmental investment and communication interact with heterogeneous consumers, fundamentalist and chartist traders, employees, a bank, a central bank, the State, assurance providers, and a capacity-constrained supervisor. The resulting system is not an empirical digital twin. It is a transparent computational laboratory in which institutional mechanisms can be altered while paired economic shocks are held constant.

The research question is: under what model conditions do capacity-constrained ex-post supervision, SME algorithmic pre-screening, and certified public environmental-data infrastructure perform differently? The question is comparative and conditional. It does not ask which institution is universally optimal. Policy performance can depend on the horizon, participation, regulatory strictness, administrative capacity, measurement failures, discounting, and the normative weights placed on accuracy, greenhushing, public expenditure, and smaller-firm burden.

The analysis isolates two institutional mechanisms. Preventive screening can correct a draft before publication but can also delay or suppress defensible communication. Certified data exchange can improve evidence only as authorization, transfer, reconciliation, and correction accumulate. An information-safe procedural architecture makes false positives, inconclusive cases, evidence conflicts, and greenhushing endogenous. The paired design maps when these mechanisms change claim quality, cost, and communication; it does not estimate real-world policy effects.

The legal architecture has a cut-off of 15 July 2026. Binding-law representations are kept distinct from policy experiments. The model represents the Unfair Commercial Practices Directive (UCPD), Directive (EU) 2024/825, corporate sustainability reporting as amended by Directive (EU) 2026/470, the EU Taxonomy, selected issuer-communication rules, the Corporate Sustainability Due Diligence Directive (CSDDD), and the European Green Bond framework. In contrast, the SME pre-screening hub and certified green data connector are proposed experimental institutions. The Commission proposal on substantiation and communication of explicit environmental claims is not treated as baseline law; its pre-verification logic is disabled by default and available only as a counterfactual.

The remainder of the paper describes the institutional map, model architecture, experimental design, findings, contributions, limitations, and archival provenance package.

2. Institutional and Regulatory Context

2.1 A layered rather than unitary legal map

The model does not posit a single EU "greenwashing code." It instead assigns claims to distinct tracks whose scope, dates, evidence requirements, and remedies differ. This separation is substantively important. A consumer-facing environmental advertisement, a mandatory sustainability report, an investor communication, a Taxonomy alignment statement, and a due-diligence failure do not share the same legal basis or penalty ceiling.

Table 1. Calendar-aware legal and institutional map

Legal cut-off: 15 July 2026. Experimental institutions are not stated as EU obligations.

Track	Model use	Date/default	Classification
UCPD	Misleading actions and omissions in B2C claims	Active throughout	Binding-law representation
Directive 2024/825	Generic claims, labels, offsets, future performance	Applies 27 Sep 2026	Binding-law representation
CSRD / Directive 2026/470	Scope, structured reporting, limited assurance	Scenario: 19 Mar 2027	Law plus configurable transposition
EU Taxonomy	Eligibility/alignment evidence and consistency	When Taxonomy claim made	Binding-law representation
Issuer communications	MAR, prospectus, transparency screens	Scope dependent	Reduced-form binding-law representation
CSDDD	Due-diligence date/remedy gate; full undertaking scope not implemented	Applies 26 Jul 2029	Reduced-form binding-law representation requiring legal review
European Green Bonds	Voluntary designation and use of proceeds	Instrument specific	Binding-law representation
Green Claims pre-verification	Disabled counterfactual	Off	Experiment, not baseline law

The UCPD supplies the general business-to-consumer baseline for misleading actions and omissions (European Parliament and Council, 2005). Directive (EU) 2024/825 amends that framework to strengthen protection against specified environmental practices, including certain generic environmental claims, sustainability labels, offset-based claims, and future environmental performance representations (European Parliament and Council, 2024a). Member States were required to transpose the Directive by 27 March 2026 and apply the measures from 27 September 2026. The simulation therefore uses 27 September 2026 as the application date for the corresponding per-se consumer rules.

Directive (EU) 2019/2161 strengthens enforcement and penalty availability for widespread infringements and widespread infringements with a Union dimension (European Parliament and Council, 2019). It requires Member States to set the maximum fine for the relevant infringements at no less than 4% of the trader's annual turnover in the Member State or Member States concerned. It does not establish 4% as a uniform EU ceiling. The unchanged model therefore treats an exact 4% cap as a LEGAL-ANCHOR scenario choice, available only when a case is classified as a coordinated widespread cross-border consumer infringement. Ordinary simulated cases use a separate 1% experimental cap applied to a simulation-scale turnover proxy.

The corporate reporting layer represents Directive (EU) 2022/2464 and the amendments in Directive (EU) 2026/470 (European Parliament and Council, 2022; 2026). Under the attached legal scenario, reporting scope requires net turnover exceeding EUR 450 million and more than 1,000 employees on average; the conditions are conjunctive and equality does not qualify. The model's configurable national-transposition scenario date is 19 March 2027. Mandatory sustainability reports receive limited assurance rather than a probabilistic audit lottery. Public enforcement selection remains separate and capacity constrained.

The EU Taxonomy is represented as an evidence and consistency track for eligibility and alignment claims, not as a firm-wide moral score (European Parliament and Council, 2020). Taxonomy-eligible and aligned turnover, capital expenditure, and operating expenditure are distinct variables. The model does not infer alignment from eligibility, and the public connector does not populate Taxonomy or net-zero fields.

Issuer and capital-market communications are screened separately under reduced-form representations of the Market Abuse Regulation, Prospectus Regulation, and Transparency Directive (European Parliament and Council, 2014; 2017; 2004). This is necessary because the UCPD is a consumer-protection instrument and is not used as a universal basis for investor-directed communication. The model also includes a voluntary European Green Bond use-of-proceeds track under Regulation (EU) 2023/2631 (European Parliament and Council, 2023a), with proceeds and expenditure ledgers that preserve the identity between issuance, unspent proceeds, and green expenditure.

The CSDDD is represented only as a reduced-form due-diligence procedure (European Parliament and Council, 2024b, as amended in 2026). Its 3% maximum is callable only for due-diligence cases, and the model's application date is 26 July 2029. Directive (EU) 2026/470 also revises the principal undertaking threshold to more than 5,000 employees and more than EUR 1.5 billion net worldwide turnover, with separate group, franchise, and third-country rules. The current model gates CSDDD by date and remedy track but does not implement that complete firm-scope test; it must therefore not be described as a doctrinally complete CSDDD implementation. Greenhushing is not modeled as an offence.

2.2 Experimental institutions

Regime B is an original EXPERIMENT institutional design, not an enacted EU pre-verification obligation. It combines the preventive substantiation and verification logic of COM(2023) 166 with RegTech and SupTech screening concepts (European Commission, 2023a; Broeders and Prenio, 2018; Financial Stability Board, 2020). The hub is a voluntary, non-binding drafting and feedback service for undertakings outside mandatory CSRD scope; the model's 1,000-employee eligibility line is a legal anchor for the experimental target group, not a general EU definition of an SME or an exemption from consumer law. Drafts are withheld during review and may be revised, withdrawn, or published, with an explicit no-endorsement disclaimer. This design tests whether early feedback corrects claims at the cost of delay or suppression of truthful communication.

Regime C is an original EXPERIMENT public data-governance infrastructure. It is not a physical emissions-monitoring device, a sensor network, a Digital Product Passport, or an enacted EU reporting obligation. Its conceptual basis is the boundary and provenance discipline of corporate GHG accounting and ESRS E1, together with interoperable data-sharing and identifier architectures in the Data Act, the Ecodesign for Sustainable Products Regulation, and the PACT methodology and technical specifications (GHG Protocol, 2004, 2011; European Commission, 2023b; European Parliament and Council, 2023b, 2024c; Partnership for Carbon Transparency, 2025a, 2025b). The connector is an interoperability and evidence-reconciliation layer: it uses data already generated by meters, suppliers, registries, and verified datasets, subject to source-specific authorization, consent and revocation. Provenance, uncertainty, coverage, staleness, mismatches, register errors, corrections, and conflicts remain explicit. It cannot supply missing underlying measurement infrastructure, and a connector record can initiate reconciliation but cannot sanction a firm by itself.

The Commission proposal COM(2023) 166 on explicit environmental claims is referenced only to delimit the counterfactual space (European Commission, 2023a). The model's pre-verification switch is off. Accordingly, no result in the baseline campaign should be interpreted as an evaluation of an enacted Green Claims Directive.

2.3 Related work and positioning

The greenwashing literature motivates the model's distinction among overstatement, incomplete evidence, selective disclosure, and communication withdrawal. Audit threat can reduce exaggerated disclosure while discouraging some truthful disclosure (Lyon and Maxwell, 2011); scrutiny and organizational norms shape selective disclosure (Marquis, Toffel, and Zhou, 2016); and greenhushing or strategic silence can limit favourable communication (Font, Elgammal, and Lamond, 2017; Carlos and Lewis, 2018). These mechanisms support endogenous revision and withdrawal, but they do not empirically calibrate either experimental institution.

RegTech and SupTech research emphasizes automated data collection and screening, prospective surveillance, supervisory capacity, data quality, explainability, and operational risk (Broeders and Prenio, 2018; Financial Stability Board, 2020). Corporate GHG standards, ESRS E1, the Data Act, Digital Product Passport architecture, and PACT specifications provide closer foundations for accounting boundaries, provenance, identifiers, and interoperable value-chain data exchange. They motivate design dimensions of the connector rather than validate its parameters or make it an implementation of an existing obligation.

Financial agent-based models are commonly assessed against fat-tailed returns, volatility clustering, and limited linear return predictability (Cont, 2001; Lux and Marchesi, 1999). Reproducing selected stylized facts is weaker than empirical calibration or out-of-sample validation. This paper therefore reports each diagnostic as reproduced, partially reproduced, or not reproduced.

Against these literatures, the comparison asks how preventive screening, interoperable evidence, and ex-post supervision differ when evidence remains uncertain and administrative capacity is scarce. The contribution is the joint mechanism test under common shocks, not a claim that the individual components are novel or that the experiment estimates policy effects.

3. Model and Methodology

3.1 Purpose, entities, and scales

The model is implemented in Python as a daily, discrete-time agent-based economy. The design follows the spirit of the Overview, Design concepts, and Details protocol for transparent agent-based model reporting (Grimm et al., 2020). Day 1 maps to 1 January 2026 by default. The model can run for 120, 365, 1,000, or 2,000 days and contains multiple listed corporate venues, heterogeneous traders, an order book, a commercial bank, a central bank, the State, consumers, workforces, assurance, and environmental-claim supervision.

Firms differ in financial state, environmental state, workforce, communication strategy, and evidence. Traders include heterogeneous fundamentalist, chartist, noise, market-making, and manipulation behaviors. Consumers are divided into segments with different attention, environmental preference, sophistication, and memory. Employees respond to internal operational signals and published supervisory outcomes. The State can pay subsidies, invest, receive sanctions, and issue European Green Bond-style instruments. Credit lines, collateral, margin enforcement, liquidation, interest, dividends, and trading frictions feed back into wealth and corporate financing.

3.2 Information-safe architecture

INFORMATION-SAFE STYLIZATION: the architecture separates the physical or latent environmental state from the information available to every operational decision-maker. This boundary is substantive: claim accuracy and supervisory performance must arise from evidence and procedure rather than oracle access.

Table 2. Information boundaries by actor

Latent truth is accessible only to the ex-post research evaluator.

Actor	Latent facts	Accessible evidence/information	Decision output
Firm	Own operations	Own records, public information	Investment, evidence, claim, qualification
Assurance provider	None	Mandatory report and submitted records	Uncertain assurance estimate/opinion
Supervisor	None	Claims, requested records, complaints, assurance, connector data	Screening, case, remedy, publication
Consumer	None	Attended claims, noisy linked evidence, decisions	Demand allocation and belief update
Investor	None	Environmental context, linked evidence by sophistication	Orders and environmental fair value
Employee	None	Noisy internal signal and public outcomes	Trust, productivity, turnover
Research evaluator	Ex-post snapshot only	Truth, intent, claims, evidence, outcomes	Research-only performance metrics

Each firm observes its own operations and may create claims and evidence. An assurance provider receives the mandatory report and the relevant records, re-performs uncertain procedures, and produces an opinion without seeing latent truth. A supervisor receives claims, submitted or requested evidence, assurance outputs, complaints, whistleblower signals, connector records, and prior cases. Consumers observe attended public claims, linked public evidence with noise, and published decisions. Investors receive an environmental context rather than a latent score; sophisticated agents can inspect linked evidence more fully than unsophisticated agents. Employees receive a noisy internal operational signal and public outcomes. Only the research evaluator accesses the private claim-to-truth mapping after the simulation path has completed. Tests in the repository are designed to prevent that mapping from entering agent decisions.

This separation resolves a common simulation problem: a regulator with direct access to truth would mechanically outperform realistic institutions, while consumers or investors using perfect information would eliminate the very asymmetry under study. Here, supervisory accuracy is an outcome of procedure and evidence, not an assumed oracle.

3.3 Environmental state, claims, and evidence

The environmental fact vector records Scope 1, 2, and 3 emissions; energy use and renewable share; water; waste and recycling; pollution; biodiversity pressure; Taxonomy-eligible and aligned turnover, capital expenditure, and operating expenditure; real environmental capital expenditure; and offsets separated from gross emissions. Records carry periods, group or operational boundaries, methodologies, and uncertainty. Scope categories and boundary metadata are motivated by GHG Protocol and ESRS E1 reporting architecture, but the vector is a STYLIZATION, not a complete compliant inventory or lifecycle assessment (GHG Protocol, 2004, 2011; European Commission, 2023b).

An environmental claim is a structured object rather than a scalar message. It identifies the firm, date, channel, audience, claim type, metric, asserted value, unit, period, boundary, linked evidence, uncertainty, qualifications, target date, offset reliance, and correction or withdrawal status. Because enforcement compares named quantities, the firm's aggregate green score is retained only for compatibility and research dashboards; it is not the sanctioning basis.

Evidence records include source, subject, period, estimate, standard error, coverage, independence, verification status, and confidence. Management evidence is neither silently converted into independent evidence nor treated as certain. The connector and limited-assurance services produce their own records with strictly positive uncertainty. The supervisor chooses relevant evidence by subject, coverage, independence, period, and accessibility.

For a metric where a larger value is environmentally better, the evidence divergence is:

$$d = \text{asserted value} - \text{evidence estimate},$$

and the one-sided standardized divergence is:

$$z\_over = \max(0, d) / \text{sqrt}(\text{evidence error}^2 + \text{stated uncertainty}^2 + \text{positive floor}).$$

The sign reverses for metrics where a lower value is greener, including emissions, pollution or water intensity, biodiversity pressure, and net-emission ratios. Materiality is the greener-than-evidence gap divided by a unit-aware scale. Default experimental bands classify z below 1 as noise; z from 1 to below 2.5 as inconclusive or requiring more evidence; statistically visible but sub-2% divergence as a correctable error; material qualified divergence as negligence; stronger material divergence as overstatement; and repeated severe findings as systemic abuse. Applicable per-se rules after 27 September 2026 can override the statistical band.

The detector includes specific safeguards. Self-declared uncertainty is plausible only up to twice the evidence standard error; excess uncertainty is disregarded and flagged. Restricted boundaries do not widen the error band unless meaningfully qualified. Repeated one-sided sub-threshold findings can escalate only after at least four findings within 365 days with a mean z-score of at least 0.5. Conflicting independent and firm records route to an inconclusive conflict process rather than a sanction. Cross-period checks address cherry-picked periods and implausible baselines. Placeholder qualifications do not count as meaningful.

3.4 Corporate choice and greenhushing

CORPORATE-CHOICE STYLIZATION: at reporting epochs, the firm searches a transparent discrete grid over real environmental investment, voluntary communication intensity, qualification, evidence effort, and possible overstatement. The objective contains a green communication premium, missed truthful premium, investment and evidence costs, qualification cost, compliance burden, and expected enforcement. The discrete grid keeps the mechanism inspectable; its coefficients and choice responses are not behaviorally estimated.

The truthful communication benchmark is:

$$q\_truthful = \text{clip}(0.20 + 0.70 \times \text{real environmental score}, 0, 1),$$

and the greenhushing gap is:

$$\text{greenhushing gap} = \max(0, q\_truthful - q\_chosen).$$

GREENHUSHING STYLIZATION: the gap compares the communication that the firm's current environmental score could truthfully support with the voluntary communication it chooses. It captures suppression relative to an internally available truthful benchmark, not an observed legal violation or a claim about managerial intent. Mandatory reporting remains separate. A firm can therefore reduce voluntary truthful communication without violating a reporting obligation; that choice eliminates the relevant voluntary claim, so the model does not invent a sanction for silence, but it reduces visibility and the truthful green premium.

The hub can alter this choice by imposing a review delay, producing warnings, encouraging meaningful revision, or prompting withdrawal. The model distinguishes prevention of good-faith errors from withdrawal-induced suppression. It reports intention-to-treat participation and treatment-on-the-treated revision measures, while corporate strategy labels remain research-only.

3.5 Assurance, supervision, and remedies

Every claim receives low-cost screening. Default experimental capacity permits 20 evidence requests and five formal investigations per reporting period, with 10% random surveillance. Complaints and whistleblowers receive higher priority; channel, claim type, evidence gap, and history contribute to the remaining risk score. Random surveillance reduces the ability of firms to game a deterministic ranking.

Cases move through recorded states: screened; evidence requested; under assessment; correction window or formal investigation; decided; published; and closed. Noise and inconclusive cases may close early. First-time correctable errors receive an experimental 30-day window, with modeled compliance at the midpoint. No monetary transfer occurs before a decision.

The ordinary simulated sanction calculation is:

$$\textit{calculated sanction} = 1.5 \times \textit{estimated benefit} + 0.02 \times \textit{affected revenue} \times \textit{severity} \times \textit{confidence}.$$

Recidivism can increase the calculated amount, after which the applied sanction is the minimum of the calculated amount, the track-specific ceiling, and corporate headroom. Simulation-scale turnover equals an experimental multiple of corporate balance; statutory annual net turnover is kept separate for legal scope tests. The 4% consumer and 3% due-diligence ceilings are track-gated. Qualification, correction, withdrawal, publication, redress, and sanctions are separate remedies.

Corrections are prospective and immutable. The original asserted value is preserved, a correction event records exposure days and case provenance, and incidence metrics remain based on the original claim. Consequently, a later correction cannot erase the historical period of misleading exposure.

3.6 Conflict resolution and the connector

Evidence disagreement is treated procedurally. A material conflict opens a CONFLICT_RESOLUTION case, enters the shared investigation queue at priority 0.85, and consumes capacity when opened. A default 20% capacity reserve prevents the conflict desk from being starved by higher-priority enforcement matters, but reduces capacity available to ordinary cases. Both quantities are experimental.

An investigation commissions a fresh measurement. Under the connector regime, the register's correction lifecycle can produce a superseding record after a delay. Otherwise a stylized third party re-measures after 30 days. The supervisor sees only the new uncertain evidence record, not physical truth. Agreement with the firm confirms the claim and corrects the external record; agreement with the register permits escalation through the ordinary evidentiary path; three-way disagreement leads to dismissal; and timeout permits confirmation, correction demand, or dismissal. Escalation without corroborated evidence is impossible by construction.

The connector authorizes each firm-source pair independently. Eight default source types cover Scope 1 and 2, renewable share, water, waste and recycling, pollution, partial Scope 3, and environmental capital expenditure. The model assumes that an underlying source record exists and operates at firm-source level rather than representing every facility, machine, subsidiary, supplier, or service-production boundary. Provenance is hashed over the payload, methodology version, period, uncertainty, completeness, and correction status, and hashes are verified before use. Data can be stale, incomplete, mismatched, unavailable, or wrong; coverage widens effective uncertainty. Reconciliation classifies matched data, rounding noise, methodological difference, incomplete coverage, source conflict, correction required, suspicious override, material overstatement, or repeated manipulation. Reconciliation itself never imposes a sanction.

3.7 Consumers, investors, employees, and financial feedback

Consumers allocate an exogenous EUR 1,000 daily budget across firms through logit utility. Segment-level environmental preferences, attention, sophistication, and memory determine the effect of claims and evidence. Accessible claim-evidence divergence reduces utility. Greenhushing reduces visibility and foregone truthful premium but does not create a controversy signal. The budget is an explicit external injection: daily gross product revenue equals the configured budget, and gross revenue is reconciled with production cost and corporate margin.

Fundamentalist investors receive an environmental context, not truth. Their environmental fair value is:

$$V\_fair = V\_fundamental \times (1 + \text{greenium gamma} \times \text{posterior score} \times \text{credibility}) \times (1 - \text{controversy discount}).$$

Sophisticated investors inspect linked evidence and respond more quickly to controversy; unsophisticated investors update more slowly. Chartists retain an indirect price-only channel. Market makers supply liquidity, other trader types place orders, and the order book determines transaction prices. Credit, collateral, margin calls, liquidation, interest, dividends, and bank exposure propagate environmental communication into financial outcomes without directly inserting the latent environmental vector into trading rules.

The direction of this financial layer must be stated precisely, because it bounds what the model can claim. Claims, evidence, and published supervisory outcomes move investor valuation and hence order flow and prices, and market prices affect corporate financing proceeds when the corporate policy sells treasury shares against the order-book midpoint. A static audit of every identified firm decision method (results/audits/) finds no other direct or indirect own-share-price input: greenwashing, disclosure, qualification, and transition-investment rules never read the firm's own price, order book, or market valuation. The equity market is therefore an endogenous market-response and financing context, not a demonstrated price-discipline mechanism, and no result in this paper relies on stock prices disciplining corporate environmental behavior.

Employees update trust from an evidence-accessible internal discrepancy and from published decisions. Quiet truthful periods allow slower recovery. Productivity affects product cost once:

$$\text{productivity multiplier} = 1 - 0.03 \times (1 - \text{trust}).$$

Annual workforce turnover begins at 10%, is capped at 30%, and is converted to a daily hazard. Departures, vacancies, hiring, 30-day onboarding, and replacement costs feed corporate accounts. The rolling 365-day employee average subsequently enters the reporting-scope test.

3.8 Event order, random streams, and accounting invariants

On each active market day the simulation: maps the legal date; updates fundamentals and environmental transition; processes prior public information through the workforce; runs product demand and books revenue and cost; creates scheduled reports and voluntary claims; performs limited assurance; screens claims and advances cases; forms information-safe investor and banking inputs; runs trading, credit, interest, dividends, and evolutionary review; and logs and validates ledgers.

This schedule prevents a claim created during the current day from affecting that day's earlier consumer demand. Same-day financial trading can respond only after a public supervisory output exists. Communications, assurance, supervision, consumer, and workforce draws use separate pseudo-random streams derived from the supervision seed. Legacy random trajectories remain unchanged when the environmental-claim module is disabled.

Accounting checks enforce single-entry pathways for corporate margin, productivity, sanctions, subsidies, evidence and replacement costs, and green-bond proceeds. A sanction received by the State equals the amount removed from corporate balances. Controversy changes demand or valuation but does not reduce the same fundamental twice.

3.9 Policy regimes

Table 3. Policy regimes compared

Regime	Institution	Primary mechanism	Legal status
A	Current EU-style supervision	Capacity-constrained screening, evidence requests, assurance, investigation, correction, sanctions	Baseline representation
B	SME algorithmic pre-screening hub	Voluntary draft review, explainable feedback, revision, withdrawal, delayed publication	Proposed experiment
C	Certified green data connector	Authorized transfers, provenance, reconciliation, corrections, conflict resolution	Proposed experiment

Regime A preserves the baseline enforcement institution and legal calendar. Regime B adds pre-publication feedback without changing the supervisor. Regime C adds certified evidence, reconciliation, and register correction while retaining ordinary supervision. Thus the paired contrast identifies the institutional mechanism plus endogenous behavioral responses under shared economic shocks. Selection into the hub and connector remains endogenous, so treatment-on-the-treated quantities are descriptive rather than causal estimates.

4. Experimental Design and Reproducibility

4.1 Paired comparison

Within each replication, all three regimes receive identical initial firms, traders, populations, strategies, latent environmental trajectories, macro shocks, market seed, and supervision seed. Only the State intervention mechanism and dedicated policy streams differ. This common-random-number design reduces irrelevant between-arm variation. Across replications and parameter draws, the stochastic environment changes.

For draw j and replication r, the attached campaign uses:

$$market\ seed = 20{,}260{,}716 + 1{,}000{,}000 \times j + 1{,}000 \times r,$$

$$supervision\ seed = 104{,}729 + 104{,}729 \times j + 7{,}919 \times r.$$

The runner explicitly re-seeds each regime, making results invariant to whether arms are executed A-B-C, C-B-A, or separately. The repository contains tests for same-seed reproducibility, order invariance, no truth leakage, correction immutability, sanction monotonicity and track gating, conservation identities, connector provenance, conflict corroboration, campaign resume behavior, and the disabled-feature legacy trajectory.

4.2 Global sensitivity campaign

Latin-hypercube sampling stratifies each parameter dimension across its range and is a standard design for computer experiments (McKay, Beckman, and Conover, 1979). The campaign samples 28 parameters classified as EXPERIMENT or STYLIZATION. Legal dates and statutory ceilings are fixed. Sampled families include the sanction-scale bridge, evidence and investigation capacity, surveillance, regulatory strictness, conflict-desk parameters, hub participation and error, connector coverage and failures, consumer response, investor sophistication, workforce trust, compliance burden, and the social discount rate.

Table 4. Attached global sensitivity campaign

Each regime run is paired within a draw-replication cell. The attached configuration records no subsets.

Horizon	LHS draws	Paired reps	Regimes	Runs	Subset
120 days	200	3	3	1,800	None
365 days	200	3	3	1,800	None
1,000 days	200	3	3	1,800	None
2,000 days	200	3	3	1,800	None
Total	800 horizon-draws	-	-	7,200	Full robustness

Each of the four horizons contains 200 parameter draws, three paired replications, and three regimes: 1,800 regime runs per horizon and 7,200 in the attached full-robustness campaign. The configuration file records master seed 20,260,716, code version, complete design matrix, parameter bounds and justifications, seed schedule, discounting, and regime identifiers. Raw draw files are written atomically and can be resumed without overwriting a valid completed draw.

The authoritative design is the machine-readable configuration in results/configuration.json: all four horizons have subset: null and all four summaries contain 200 draws. The current reproducibility guide and data dictionary have been synchronized to this layout. Historical selective-confirmation commands are not part of the reported design.

4.3 Outcomes and inference

The evaluator distinguishes research-only truth-based metrics from quantities observable to in-model institutions. Research-only outcomes include original material overstatements, exposure-weighted severity, population detection recall, severity-weighted recall, strategic versus measurement-noise incidence, and withheld truthful claims. They are calculated after a run and never supplied to agents.

Observable or procedural outcomes include backlog, queue age, case completion, conflict cases, correction delay, hub participation, meaningful revisions, noise flags, greenhushing, public and firm cost, employee trust, replacement cost, investment, and final emissions. Original incidence is based on claims as published; live-uncorrected incidence recognizes subsequent correction; exposure-weighted severity multiplies harm by exposure duration; misleading-claim days preserve the temporal burden of delayed action.

For each draw, paired mean differences are calculated across the three replications. Effect distributions then summarize 200 draw-level paired effects. For harmful outcomes and costs, signs are normalized so that a positive effect means improvement relative to current EU supervision. Thus a negative greenhushing or public-cost effect means that the policy increased the gap or cost. The summaries report means, standard deviations, 95% interval half-widths across draws, quantiles, and the fraction of draws with improvement. These are model-sensitivity summaries, not sampling-based estimates of a real population.

Composite rankings use five discrete experimental weight scenarios: default, accuracy-focused, anti-greenhushing, cost-focused, and SME-protective. Within each draw, each criterion is min-max normalized across the three regimes before the weighted values are summed. The resulting score is therefore a relative ranking device, not an absolute welfare function; its construction and weights are normative EXPERIMENT choices (OECD and European Commission, Joint Research Centre, 2008). Raw metrics, four-axis Pareto membership, and rank frequency are reported alongside it. Standardized regression coefficients and partial rank correlation coefficients summarize parameter dependence, but these associational diagnostics can miss interactions and nonlinearity (Marino et al., 2008; Pianosi et al., 2016).

4.4 Replication robustness extension

Three paired replications per LHS draw are appropriate for broad global screening but can under-sample rare escalation, institutional failure, and tail outcomes. A separate dry-run-first workflow deterministically selects 12 configurations by maximin coverage of standardized cross-horizon outcomes and targets 15 paired replications by default (configurable from 10 to 20). It preserves the authoritative first three replications and adds only replications 4-15 under the same paired seed schedule.

The extension has been executed. Draws 41, 42, 69, 71, 75, 104, 109, 136, 144, 147, 180, and 187 were extended to 15 paired replications at each horizon, and all 48 draw-horizon extension files pass validation. Moving from three to 15 replications narrows the paired 95% interval in 92.3% of the 1,743 draw-metric-policy cells with a finite comparison, with a median half-width ratio of 0.26. The default-weight winner computed at three replications is confirmed at 15 replications in 32 of 48 draw-horizon cells (9, 9, 8, and 6 of 12 at 120, 365, 1,000, and 2,000 days), and the complete three-regime ranking is confirmed in 60.4% of cells. Draw-level winner labels under three replications are therefore noisy, especially at long horizons; the paper accordingly emphasizes distributional effects, improvement fractions, and Pareto membership across the 200 draws rather than draw-level league tables. The first three replications remain the authoritative campaign, and extension outputs are stored separately in results/replication_robustness/.

5. Findings

5.1 Conditional policy maps

Ranking instability explains why the two policy mechanisms must be reported conditionally rather than collapsed into a universal league table. The default-weight winner changes across horizons for 161 of 200 matched parameter draws (80.5%). Within a horizon, changing normative weights changes the winner in 70.0% of 120-day draws, 63.0% of 365-day draws, 58.0% of 1,000-day draws, and 45.5% of 2,000-day draws. These reversals are consequences of distinct accuracy, greenhushing, cost, and timing trade-offs, not evidence that comparison is impossible.

Table 5. Default-weight winner shares and within-horizon weight sensitivity

Shares are counts divided by 200 LHS draws. They are not empirical success probabilities.

Horizon	Current supervision	Pre-screening hub	Data connector	Weight-sensitive draws
120	47.5%	48.0%	4.5%	70.0%
365	53.0%	38.0%	9.0%	63.0%
1,000	35.5%	13.5%	51.0%	58.0%
2,000	33.5%	7.0%	59.5%	45.5%

At 120 days, current supervision and the hub are nearly tied under default weights, winning 47.5% and 48.0% of draws; the connector wins 4.5%. At 365 days, current supervision leads at 53.0%, followed by the hub at 38.0% and the connector at 9.0%. The ordering changes at long horizons: the connector wins 51.0% at 1,000 days and 59.5% at 2,000 days, while the hub falls to 13.5% and 7.0%. The hub does not cease to improve claim-quality metrics; it loses default-weight competitiveness because review and revision costs recur, participation is selective, and greenhushing is penalized, while connector evidence accumulates. Current supervision remains competitive because cost and greenhushing enter the composite. These frequencies are not probabilities that a policy will succeed in the European economy. They are shares of the model's 200 stratified parameter configurations.

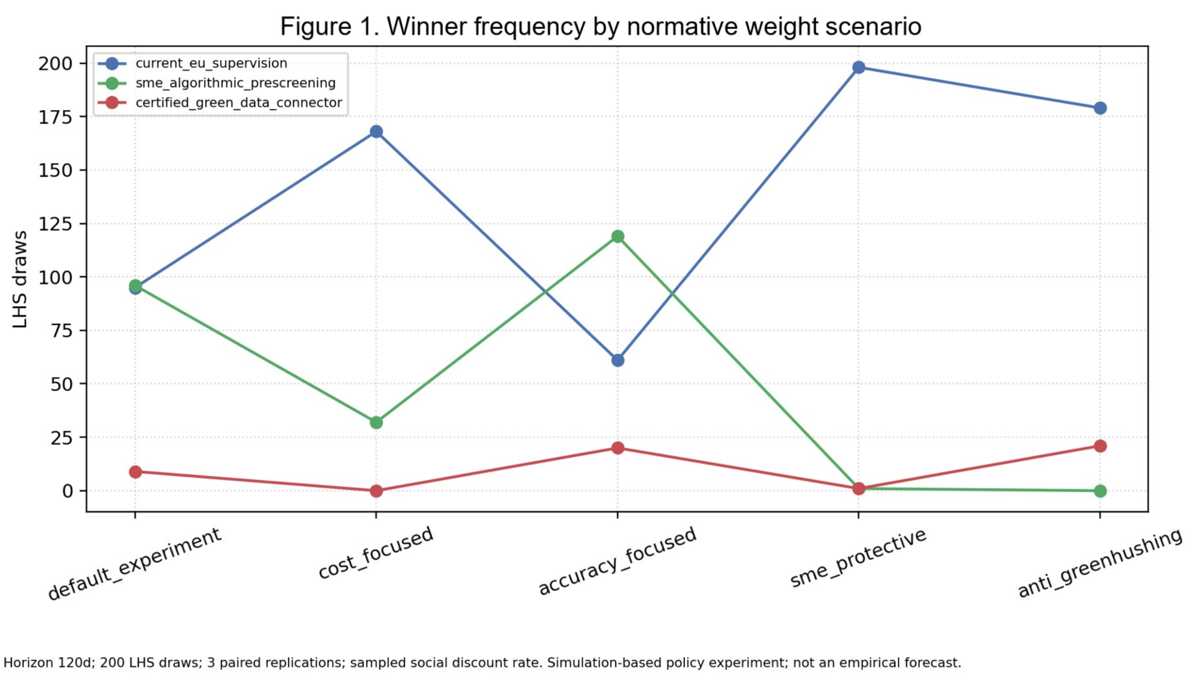


Figure 1. Winner frequency by normative weight scenario at 120 days.

Source: attached simulation output. Counts are across 200 LHS draws with three paired replications; not an empirical forecast.

The five weight scenarios explain why no regime dominates. At 120 days, the accuracy-focused scenario selects the hub in 119 of 200 draws, while the cost-focused and anti-greenhushing scenarios select current supervision in 168 and 179 draws. Under the SME-protective weights, current supervision wins 198 draws because the experimental hub adds burden and the connector adds cost. In this model, a regime lies on the four-axis Pareto frontier when no alternative is at least as good on exposure severity, public cost, greenhushing, and accuracy and strictly better on at least one of those criteria. Current supervision is on that frontier in all 200 draws at every horizon; the hub appears in 183, 180, 180, and 184 draws; and the connector appears in 164, 174, 200, and 200 draws at 120, 365, 1,000, and 2,000 days. A high Pareto frequency means that a regime is not strictly dominated on the selected criteria, not that it is preferred under all values or that it dominates the other regimes.

5.2 Pre-screening: lower modeled overstatement, higher greenhushing

The hub reduces original material overstatement relative to current supervision in 91.0% of 120-day draws, 88.5% at 365 days, 87.5% at 1,000 days, and 88.0% at 2,000 days. It reduces exposure-weighted severity in 85.0%, 90.5%, 89.0%, and 87.5% of draws. Mean paired improvements in the original material count are 8.65, 13.74, 24.20, and 45.99 claims as horizons lengthen. Because claim volume and exposure accumulate over time, these absolute effects should not be compared across horizons as rates.

Table 6. Principal paired effects relative to current EU-style supervision

Positive probability columns mean improvement. Negative delta greenhushing/cost values mean higher gap/cost. Costs are simulation-scale EUR.

Days	Regime	P(improve) original	P(improve) severity	Mean delta greenhushing	Mean delta public cost
120	Hub	91.0%	85.0%	-0.00462	-EUR 24,290
120	Connector	50.5%	40.5%	-0.000001	-EUR 64,071
365	Hub	88.5%	90.5%	-0.00637	-EUR 26,462
365	Connector	68.5%	60.5%	-0.00013	-EUR 70,352
1,000	Hub	87.5%	89.0%	-0.00574	-EUR 28,751
1,000	Connector	100.0%	97.0%	-0.00035	-EUR 86,192
2,000	Hub	88.0%	87.5%	-0.00502	-EUR 32,088
2,000	Connector	100.0%	100.0%	-0.00044	-EUR 111,608

The model does not justify calling all prevented publication fraud prevention. The hub detects missing units, periods, boundaries, qualifications, evidence, and uncertainty, and can prompt meaningful correction of good-faith errors. It also delays publication and can induce withdrawal. The honest interpretation is therefore prevention through a mixture of revision and claim suppression. Strategy labels and the ex-post intent snapshot help separate mechanisms for research, but selection into the voluntary hub remains endogenous.

The trade-off is visible in greenhushing. The normalized paired effect is negative at every horizon: mean changes of -0.00462, -0.00637, -0.00574, and -0.00502 indicate an increased greenhushing gap relative to current supervision. The hub improves greenhushing in only 0%, 3.0%, 4.5%, and 6.5% of draws. It also increases discounted public cost in every draw, with mean paired cost effects of -EUR 24,290, -EUR 26,462, -EUR 28,751, and -EUR 32,088. These are simulation-scale euros and should not be mapped to an EU budget.

Sensitivity results reinforce the mechanism. At 120 days, the largest partial rank correlations with the hub's severity improvement are participation scale (PRCC 0.531) and strictness (0.446); at 365 days they are 0.552 and 0.494. The same parameters have negative associations with the greenhushing effect: at 365 days, participation PRCC is -0.550 and strictness is -0.496. More participation and stricter screening improve modeled claim quality while worsening communication suppression. This is a structural trade-off within the model, not an externally validated elasticity.

5.3 Connector: horizon-dependent benefits and persistent cost

The connector has limited short-horizon influence because authorization, transfer, reconciliation, correction, and conflict resolution take time. At 120 days it improves exposure-weighted severity in 40.5% of draws, with a mean normalized effect of -0.63, essentially centered around no improvement relative to the cross-draw dispersion. At 365 days it improves severity in 60.5% of draws and the mean paired effect becomes 44.61. At 1,000 and 2,000 days, the improvement shares rise to 97.0% and 100%, with mean paired effects of 1,855.23 and 12,216.10 accumulated exposure-severity units. Original material overstatements improve in 50.5%, 68.5%, 100%, and 100% of draws across the four horizons.

These full-space long-horizon findings are stronger than the earlier subset results described in the documentation. They remain model results under the sampled ranges. The connector's long-run claim-quality advantage arises from recurring evidence transfers, accumulated provenance, repeated comparison, correction, and better population detection. Its one-time setup and per-firm integration costs become a smaller share of the accumulated comparison at longer horizons, while daily operating and governance costs continue; the model applies no accounting depreciation schedule. This does not demonstrate that a real public data connector would be effective or cost-beneficial.

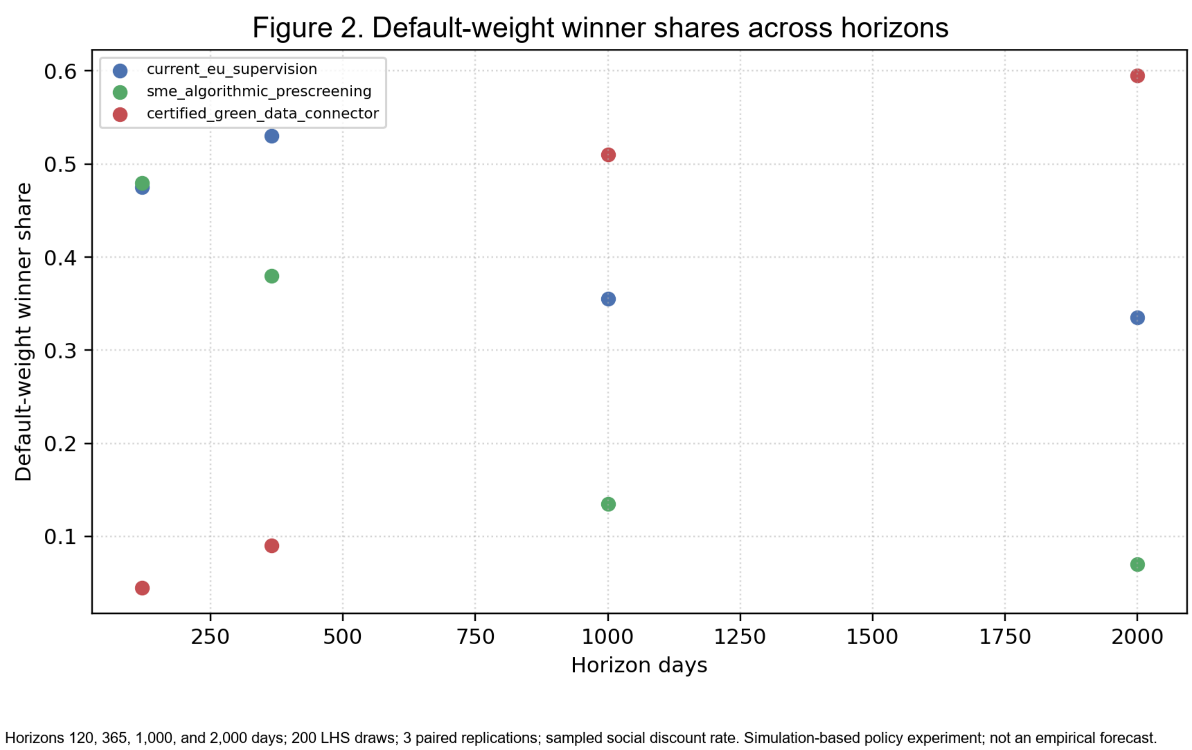


Figure 2. Default-weight winner shares across horizons.

Source: attached full-robustness campaign. Each horizon contains 200 LHS draws and three paired replications.

The cost result remains adverse. The connector increases discounted public cost in every draw, with mean normalized effects of -EUR 64,071, -EUR 70,352, -EUR 86,192, and -EUR 111,608. Social discount rate is the strongest parameter associated with the connector's cost effect, with PRCC values of 0.756, 0.688, 0.716, and 0.839 across the four horizons. The connector has a near-zero mean greenhushing effect, but the improvement fraction remains only 12.0%, 12.0%, 13.0%, and 10.5%. This should be interpreted as little systematic help on the greenhushing criterion, not as evidence of large suppression.

Connector severity effects become more sensitive to economy-wide behavioral settings over time. At 1,000 and 2,000 days, regulatory strictness has PRCC values of -0.616 and -0.647 with connector severity improvement, while compliance burden has -0.490 and -0.530. The result reflects interactions between baseline communication choices and the marginal information supplied by the connector. These coefficients are associational sensitivity diagnostics.

5.4 Conflict desk and source errors

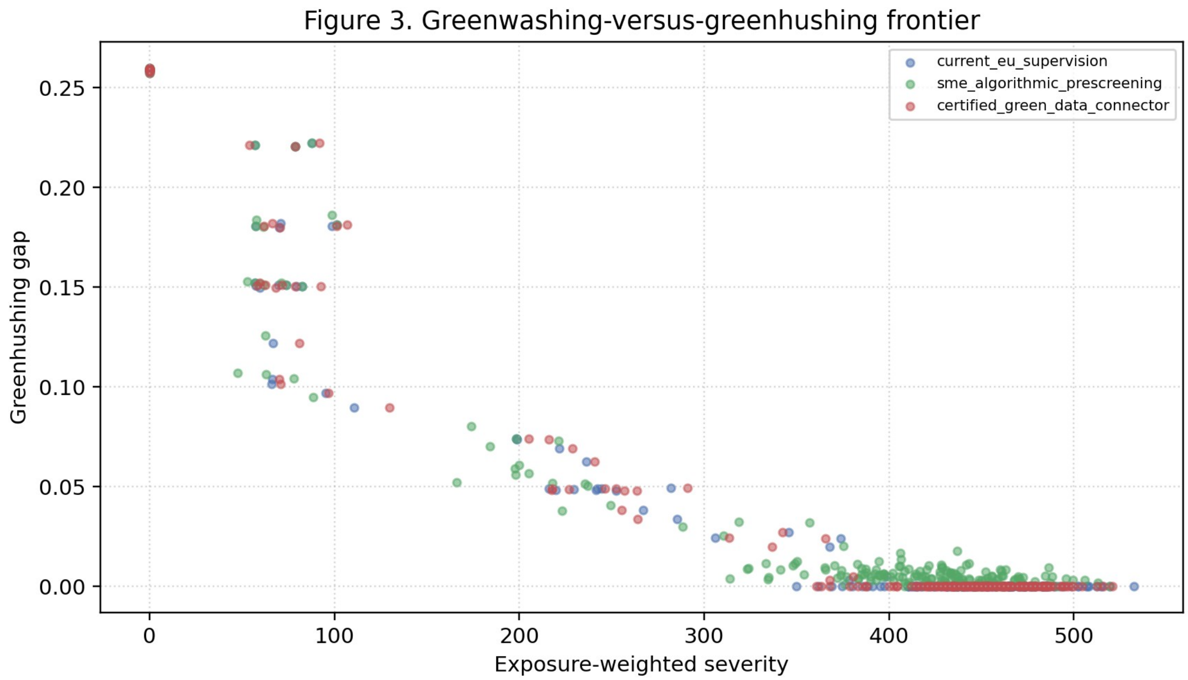
Connector data create genuine disputes that the baseline encounters less often. Averaged over the 600 regime runs per horizon, the connector opens 0.87 conflict cases at 120 days, 2.28 at 365 days, 4.50 at 1,000 days, and 7.99 at 2,000 days. It consumes 0.71, 1.86, 3.87, and 6.88 investigation slots, respectively. At 365 days, neither current supervision nor the hub opens a conflict in the campaign mean; at 1,000 days their means are 0.71 and 0.79 cases, compared with 4.50 for the connector. At 2,000 days they open 2.08 and 2.10, compared with 7.99.

Resolution is slower under the connector because corrections and re-verification are real processes. Mean connector resolution delay is 44.43 days at 365 days, 75.98 days at 1,000 days, and 102.47 days at 2,000 days. Relative to baseline, the normalized delay effects are -44.43, -58.27, and -46.64 days. At 120 days, open disputes have generally not matured into resolved-delay observations. The capacity reserve prevents indefinite starvation but competes with ordinary enforcement. Higher source error loads therefore burden the dispute channel.

The safeguard is that source error does not become a direct sanction channel. A register mismatch produces a conflict; a material case can escalate only after independent corroboration. At 2,000 days, connector runs average 1.18 corroborated escalations and 2.79 register corrections, compared with 0.36 escalations and 1.40 register corrections under current supervision. These figures show procedural activity, not evidence of real-world error rates. The stress tests described in the repository use high register-error scenarios to verify that unresolved conflict cannot support a monetary sanction.

5.5 Greenwashing-greenhushing frontier and real-economy interpretation

The simulated policy problem is a frontier rather than a one-dimensional detection contest. Draws with lower exposure-weighted severity can have larger greenhushing gaps, and draws with extensive truthful communication can expose more claims to delayed enforcement. Figure 3 plots this frontier for the 120-day campaign.



Horizon 120d; 200 LHS draws; 3 paired replications; sampled social discount rate. Simulation-based policy experiment; not an empirical forecast.

Figure 3. Greenwashing-greenhushing frontier at 120 days.

Source: attached simulation output. Exposure-weighted severity and the greenhushing gap are model quantities.

The real-economy channels give these communication outcomes economic meaning: consumers can misallocate the fixed daily budget; investors can distort price around environmental signals; and employees can lose trust, reduce productivity, or leave. However, the directions and magnitudes of these mappings are stylizations. The campaign does not establish causal demand, price, trust, productivity, or turnover effects in observed data. Consequently, policy rankings should be reported using the raw procedural and claim-quality metrics alongside composite economic scores. Because firm communication and investment rules contain no own-share-price feedback (Section 3.7), investor signal distortion is a market-response metric rather than evidence of price discipline.

5.6 Summary of robust and non-robust findings

Table 7. Robustness interpretation

Finding	Evidence in attached campaign	Permissible interpretation
Hub lowers claim harm	Original and severity improve in about 85-91% of draws across horizons	Robust within sampled model ranges; mechanism includes revision and withdrawal
Hub raises greenhushing	Greenhushing improves in only 0-6.5% of draws; public cost worsens in all draws	Prevention has a communication-suppression and fiscal trade-off
Connector is horizon dependent	Severity improves in 40.5%, 60.5%, 97%, and 100% of draws	Short-horizon null-to-mixed effect; stronger accumulated long-horizon effect
Connector is costly	Discounted public cost worsens in all draws at every horizon	Long-run claim-quality gains do not imply simulated cost savings
Rankings are unstable	80.5% cross-horizon reversal; 45.5-70% weight-sensitive within horizon	No universal winner or unconditional policy recommendation

Finding	Evidence in attached campaign	Permissible interpretation
Conflict safeguards matter	Connector opens and investigates more disputes; escalation requires corroboration	Source errors load procedure rather than directly producing sanctions

The most robust within-model statement is that the hub improves claim-quality metrics in a large majority of parameter draws while increasing greenhushing and public cost. A second robust statement is that the connector is highly horizon-dependent and cost-intensive. The least defensible statement would be that one regime is universally optimal. Ranking instability, Pareto co-membership, and sensitivity to normative weights directly contradict such a claim.

5.7 Financial-market validation

The market engine is validated in a dedicated fixed-parameter campaign, separate from the policy LHS: 30 independently seeded 10,000-day paths under the baseline current-supervision regime, each discarding a 2,000-day burn-in and exporting the daily market series and the primary listing's complete order-book event stream (about 256,000-298,000 events per seed). Every seed carries a provenance manifest recording the full configuration, independent market and supervision seeds, schema versions, and content hashes. Diagnostics are computed per seed under rules declared before execution, and cross-seed statements use the seed as the unit of inference; raw returns are never pooled across paths. An earlier illustrative diagnostic of the detached single-path simulation_results.csv is superseded by this campaign.

Table 8. Multi-seed financial stylized-facts diagnostics (30 independent seeds)

Source: results/financial_validation/pilot_30seeds/diagnostics/. Statuses follow rules declared before execution; the seed is the unit of inference. R/P/N = seeds classified reproduced / partially reproduced / not reproduced. No simulation was executed during manuscript revision.

Fact	Cross-seed diagnostic (means over 30 seeds)	R/P/N	Status
Volatility clustering	Absolute-return ACF 0.170 and squared-return ACF 0.183 at lag 1, beyond the 95% band at multiple lags	30/0/0	Reproduced
Positive spread and depth	Mean spread 1.38 ticks; two-sided depth in essentially all snapshots	30/0/0	Reproduced
Trade-sign persistence	Lag-1 trade-sign ACF 0.486 against an approximate 0.009 white-noise band	30/0/0	Reproduced
Cancellation activity	Cancellation-to-submission ratio 0.537	30/0/0	Reproduced
Volume-volatility relation	Mean daily volume-volatility correlation 0.113	17/13/0	Partially reproduced
Fat-tailed returns	Mean excess kurtosis 1.99 (cross-seed SD 1.33); mean Hill alpha 5.38	3/27/0	Partially reproduced
Weak linear return autocorrelation	Mean lag-1 return ACF -0.082, small but outside the white-noise band	0/29/1	Partially reproduced

Fact	Cross-seed diagnostic (means over 30 seeds)	R/P/N	Status
Positive price impact	Mean correlation of log volume with immediate absolute mid-price movement -0.013 (95% half-width 0.004)	0/4/26	Not reproduced

Four microstructure and volatility facts are reproduced in every seed. Two return-distribution facts are borderline. Excess kurtosis is positive in all seeds but exceeds 3 in only a minority, and Hill tail exponents average 5.4, so simulated tails are heavier than Gaussian yet thinner than the empirical 2-5 reference band in most seeds. Linear return autocorrelation is small but negative beyond the white-noise band at short lags, so the weak-autocorrelation criterion is only partially met. The multi-seed result also reverses the earlier single-path impression that fat tails were strong and linear predictability was severe; that detached path was unrepresentative, which is precisely why cross-seed validation is required.

Positive price impact is not reproduced. The implemented diagnostic correlates log trade volume with the immediate absolute mid-price movement; across seeds the mean correlation is -0.013 with a cross-seed 95% half-width of 0.004, classified as not reproduced in 26 of 30 seeds and never as reproduced. Under this unsigned immediate-horizon measure, the engine's mid-price does not respond to trade size in the way empirical impact regularities require. A signed event-level diagnostic - trade sign multiplied by the future mid-price change over multiple event horizons, with seed-level uncertainty - is the sharper standard test and is identified as future work; under the pre-declared rule the fact is currently not reproduced and the paper reports it as such.

These results bound interpretation rather than invalidate the policy comparison. The policy outcomes in Sections 5.1-5.6 are claim-quality, greenhushing, procedural, and cost metrics generated by enforcement, information, and capacity mechanisms. As established in Section 3.7, firm environmental decisions contain no own-share-price feedback, so no policy ranking depends on price-mediated corporate incentives. What the partial validation does restrict is the reading of market-side quantities: investor signal distortion and price paths are internal model responses, and the financial market must not be described as empirically validated.

6. Contributions and Novelty

The central contribution is a mechanism-based comparison of ex-post supervision, preventive drafting support, and certified environmental-data exchange within one financial and real-economy simulation. The comparison shows how the timing and information architecture of an institution can change claim quality, communication, cost, and administrative workload without treating any experimental regime as enacted law. The financial market is an endogenous response and financing layer, not a source of own-share-price discipline over corporate environmental choices.

The information-safe procedural design is essential to that comparison. Supervisors and assurance providers receive uncertain records rather than latent truth; cases proceed through screening, evidence requests, correction, investigation, decision, publication, and closure; and conflicts consume capacity without supporting sanctions unless evidence is corroborated. This makes false positives, inconclusive cases, exposure duration, and greenhushing outcomes of institutional procedure rather than assumptions away from the analysis.

The paired global design then tests these mechanisms across uncertain environments and normative choices. Raw effects, Pareto membership, rank frequency, parameter sensitivity, and alternative weights constrain interpretation of the composite score. The LEGAL, LEGAL-ANCHOR, EXPERIMENT, and STYLIZATION registry makes the source and role of assumptions visible without implying empirical calibration.

7. Limitations

Six limitations bound the interpretation of the experiment.

First, the model is neither empirically calibrated nor a source of causal inference. Enforcement frequencies, claim prevalence, behavioral responses, emissions, administrative costs, and the mappings from claims to demand, valuation, trust, productivity, and turnover are not estimated from observed data. The LHS campaign tests robustness to chosen ranges rather than validating them; paired common shocks identify differences only within the simulation. Voluntary participation is endogenous, so intention-to-treat and treatment-on-the-treated quantities remain descriptive. The outputs are not real-world forecasts.

Second, legal and institutional implementation is reduced-form and scenario-specific. National transposition, jurisdiction, appeals, remedies, enforcement cooperation, complete CSDDD undertaking scope, the full ESRS datapoint architecture, Taxonomy technical criteria, lifecycle assessment, assurance standards, and European Green Bond workflows require separate legal and technical treatment. The connector also abstracts from the engineering, governance, lawful-basis, cybersecurity, and organizational requirements of GDPR and NIS2. The paper is not legal advice.

Third, measurement interoperability and environmental coverage are partial. The model detects unit mismatch but does not represent conversion evidence; Scope 3, facilities, subsidiaries, individual machines, service-production boundaries, and supply chains are incomplete. Methodology changes, boundary differences, staleness, and missing coverage enter through stylized uncertainty. The detector also cannot resolve production-driven absolute reductions, intensity-versus-absolute framing without production evidence, coordinated narratives, or audience perceptions of technically true claims. The connector reconciles available records but cannot create missing meters, supplier data, or a complete environmental truth system. Claim-quality outcomes must therefore be distinguished from physical-emissions outcomes.

Fourth, the composite score and its weights are normative EXPERIMENT aggregations, not an absolute welfare function. Default winner shares describe one value system; alternative weights and frequent Pareto co-membership show that rankings depend on accuracy, greenhushing, cost, participation, and burden priorities. The exogenous EUR 1,000 consumer budget, simulation-scale corporate balances, and simulation euros are accounting devices and cannot be read as EU fiscal costs or firm forecasts.

Fifth, statistical and computational reproducibility are limited. The authoritative campaign uses three paired replications per draw, so rare events and draw-level rankings can remain noisy. The executed 15-replication extension on 12 representative configurations narrows paired intervals in 92.3% of comparable cells yet confirms the three-replication default-weight winner in only 32 of 48 draw-horizon cells; distributional effects, improvement fractions, and Pareto membership are therefore more reliable than draw-level league tables. Partial rank correlations and standardized coefficients can miss non-monotonicity and interactions. In addition, the shipped campaign manifests record a git hash suffixed +dirty. The archival provenance package preserves the authoritative outputs, configuration, seed schedules, recoverable environment information, and SHA-256 checksums, but the original dirty diff was not retained. A later direct-child commit is only a partially reconstructed candidate source snapshot; recovery of the original dirty worktree and bit-level reproducibility are not claimed.

Sixth, financial-market validation is limited and the model contains no own-share-price feedback into corporate greenwashing, disclosure, or transition investment. Across 30 independent seeds, volatility clustering, positive spread and depth, trade-sign persistence, and cancellation activity are reproduced; fat tails, the volume-volatility relation, and weak linear return autocorrelation are only partially reproduced; and positive price impact is not reproduced under the implemented immediate-impact diagnostic. Market-side quantities are therefore internal responses, and no policy ranking is evidence of equity-price discipline over corporate environmental behavior.

8. Conclusion

Within the experiment, preventive pre-screening frequently lowers material overstatement and exposure-weighted severity because it acts before publication, but recurrent review and revision costs, selective participation, delay, and withdrawal raise greenhushing and reduce its default-weight competitiveness over longer horizons. The hub therefore continues to improve claim quality in a large share of draws; it does not simply fail.

The connector starts slowly because authorization, transfers, reconciliation, correction, and conflict resolution take time. Its long-horizon claim-quality advantage emerges from recurring evidence, accumulated provenance, repeated comparison, and shorter exposure to inaccurate claims. It becomes the most frequent default-weight winner at 1,000 and 2,000 days, but remains costly, generates conflicts, and never Pareto-dominates current supervision. The 80.5% cross-horizon reversal rate therefore supports conditional comparison: rankings depend on horizon, participation, evidence quality, administrative capacity, discounting, costs, and normative objectives.

The model is a computational laboratory rather than a policy recommendation engine, empirical forecast, or causal evaluation. Its conclusions concern mechanisms under explicit EXPERIMENT and STYLIZATION assumptions. Future work should calibrate behavioral, enforcement, measurement, and cost channels; extend facility and value-chain representation; increase stochastic replication; and develop country-specific legal modules. Financial-market extensions should first repair the price-impact diagnostic and would require a separately pre-registered, information-safe own-share-price feedback channel if market discipline became the research question.

Data and Code Availability

The complete model, parameter registry, tests, run-level outputs, manifests, figures, and reproducibility instructions are contained in the attached repository. The archival provenance package in provenance/global_lhs_2026-07-15/ documents the global LHS campaign's recorded code version, 1645f3589408503894c7e8f44a92ffe382a9a5b4+dirty, together with the documented base commit, a partially reconstructed candidate source snapshot, execution configuration, recoverable environment information, limitations, and SHA-256 checksums. It preserves the original outputs unchanged but does not claim recovery of the original dirty worktree or bitwise reproducibility.

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Appendix A. Parameter Classification and Selected Ranges

The registry defines four role classes. LEGAL values implement binding dates, conjunctive thresholds, or track-specific ceilings. LEGAL-ANCHOR values use a legal boundary to define an experimental target population but do not create an exemption. EXPERIMENT values specify institutional mechanisms or evaluation choices. STYLIZATION values encode behavioral, accounting, or scale relationships. Evidence classes separately identify legally mandated, reference-class, or illustrative-scenario values.

Table 9. Parameter-role taxonomy and sampled families

Detailed defaults, bounds, code locations, and justifications appear in docs/PARAMETER_REGISTRY.md.

Family	Examples	Role class	Selected LHS ranges
Legal calendar	Application dates, CSRD thresholds, gated ceilings	LEGAL	Fixed; not sampled
Target eligibility	Hub 1,000-employee line	LEGAL-ANCHOR	Fixed; not sampled
Sanctions/capacity	Benefit factor, evidence and investigation slots, surveillance	EXPERIMENT / STYLIZATION	0.5-3x benefit; 5-40 requests; 1-12 investigations
Conflict desk	Reserve share, resolution days	EXPERIMENT	0-0.60 reserve; 5-60 days
Hub	Participation, strictness, noise, delay	EXPERIMENT	0.2-1.5; 0.1-0.95; 0-0.50; 0-20 days
Connector	Coverage, mismatch, register error, staleness, downtime	EXPERIMENT	Coverage 0.5-1.2; failure probabilities 0-0.25
Behavior	Consumer, investor, workforce, compliance burden	STYLIZATION	Documented registry ranges; no empirical calibration
Evaluation	Social discount rate, score weights	EXPERIMENT	0-7% per year; five discrete weight scenarios

All 28 globally sampled dimensions are EXPERIMENT or STYLIZATION. Examples include evidence-request capacity of 5-40 cases per period; investigation capacity of 1-12; random surveillance of 0-0.30; correction windows of 10-90 days; hub strictness of 0.10-0.95 and noise of 0-0.50; connector mismatch of 0-0.10, register error of 0-0.05, staleness of 0-0.25, downtime of 0-0.10, and correction delay of 10-120 days; consumer preference scale of 0.5-1.5; investor sophistication of 0.05-0.80; workforce trust loss of 0.10-1.00; compliance burden scale of 0.25-2.50; and social discount rate of 0-0.07 per year. These ranges define the computational experiment and are not confidence intervals around estimated parameters.

Appendix B. Outcome Definitions

Original material overstatements. Count of claims whose original published value exceeds ex-post truth by at least the evaluator's experimental materiality threshold. This is research-only.

Live-uncorrected material overstatements. Material claims still active after prospective correction and withdrawal events.

Exposure-weighted severity. Claim severity multiplied by exposure days, preserving harm from delayed correction. A discounted counterpart is used in comparisons while the undiscounted ledger is retained.

Population detection recall. Share of all original material claims that receive a relevant detected outcome. Unlike screening-conditioned recall, the denominator includes claims never selected because of limited capacity. This is research-only.

Greenhushing gap. Positive difference between the truthful communication benchmark and chosen voluntary communication intensity, averaged over relevant observations.

Meaningful pre-publication revision. Revision that addresses a legally material hub issue. Informational or noise flags do not count as prevention.

Conflict metrics. Cases opened, investigations consuming capacity, unresolved cases, resolution delay, firm confirmation, register correction, claim correction, dismissal, and corroborated escalation.

Discounted total public cost. State policy and regulator time cost discounted at the draw-specific social rate. It is a simulation-scale accounting metric.

Pareto membership. Non-dominance on lower severity, lower public cost, lower greenhushing, and higher accuracy. It is reported alongside, not replaced by, composite ranks.

Appendix C. Reproducibility and Seed Strategy

The campaign design and seed schedule are deterministic conditional on the archived source and execution environment.

The top-level publication driver refuses fewer than 200 draws, three paired replications, or 120 days unless explicitly placed in development mode. Re-aggregation can be performed without re-running simulations. The attached configuration specifies run_global_lhs_campaign.py --full-robustness. The reproducibility guide provides commands for the historical subset workflow as well as the full driver. All outputs should be interpreted with the machine-readable manifest corresponding to the reported results and with the archival provenance package. The latter records the original +dirty label and does not claim bit-level reproducibility of the original working tree.

Appendix D. Additional Robustness Interpretation

At 120 days, default winner counts are 95 for current supervision, 96 for the hub, and nine for the connector. At 365 days they are 106, 76, and 18. At 1,000 days they are 71, 27, and 102. At 2,000 days they are 67, 14, and 119. Across all horizons, the anti-greenhushing and cost-focused scenarios favor current supervision more strongly than the default scenario.

The decline in weight sensitivity from 70.0% to 45.5% as the horizon lengthens does not establish convergence to one universal policy. The connector's default winner share rises, but current supervision remains the modal choice under cost-focused and SME-protective weights at 2,000 days. The hub's persistent severity improvements are offset by its greenhushing and cost effects. Accordingly, the model supports conditional policy maps and Pareto comparisons rather than a scalar league table.